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FRACTIONAL-ORDER NEWTON–RAPHSON METHOD FOR NONLINEAR EQUATION WITH CONVERGENCE AND STABILITY ANALYSES

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Abstract

Fractional-order techniques have many applications in real-life problems nowadays. The utilization of fragmentary math in many parts of science and engineering is wide and somewhat recent. There are various types of subsidiaries that can be valuable in various issues. In this paper, we focus on the effect of this kind of fractional derivative in the search for roots of nonlinear equations and its dependence on the initial estimations. We will check the convergence and stability analyses of the fractional Newton–Raphson (FNR) method for the proposed definitions. We will apply fractional Riemann–Liouville and Caputo-derivatives in the standard Newton root-finding method for different examples and also verify convergence and stability for these problems. Finding a root is very important for nonlinear equations which are used to solve chemical, biological, and engineering problems.

Keywords: Roots; Nonlinear Equations; Fractional Operators; Stability; Convergence Analysis.

1. INTRODUCTION

In mathematical analysis, finding the solution to the problem with an algorithm with the help of numerical approximation is known as numerical analysis. Numerical analysis is used in many real-life problems. Before the 21st century, it was only used in engineering and physical sciences but after that, its applications have been seen in the field of social science, business, and medicine as well as in scientific computation. Numerical methods mostly depend on hand interpolation formulas for large data analysis before the advent of modern computers. Till the mid-twentieth century, PC can figure out the necessary capacities yet a considerable number of equations cannot be utilized with the assistance of programming calculations.[1]

The Newton–Raphson method (also known as Newton's method) is a way to quickly find a good approximation for the root of a real-valued function $f(x) = 0$. It is named after Isaac Newton and Joseph Raphson and is a root-discovering calculation that creates progressively better approximations to the roots (or zeroes) of a genuinely esteemed capacity. It utilizes the possibility that a nonstop and differentiable capacity can be approximated by a straight-line digression to it. At first, Newton applied the technique just to polynomials which start with a beginning root gauge and separate a succession of mistake adjustments. He utilized every rectification to modify the polynomial as far as the excess, and afterward tackled another revision by dismissing more serious level terms.[2] Newton's strategy was first distributed in 1685 in A Composition of Polynomial math both Authentic and Commonsense by John Wallis.[3]

Recently, many scientists have demonstrated that fractional models describe natural phenomena in an accurate, systematic way better than the classic integer-order counterparts with ordinary time derivatives.[4,5] The fractional-order models give better results than the real data. Some fractional operators are given in Refs. 6 and 7, and applications of these fractional operators are given in Refs. 8 and 9.

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The investigations of some other infectious disease mathematical models have been studied in Refs. 10–14.

In this work, we find the roots of the nonlinear equation by using the fractional-order Newton method which is very important for real-life problems to find convergence solitons. Convergence and stability analyses were also discussed in Secs. 3–5 with different applications. Numerical simulations were also carried out to show the effect of the fractional technique on the problems in Sec. 6. In the end, the importance and effect of these results explain in the conclusion of Sec. 7.

2. BASIC CONCEPTS OF FRACTIONAL OPERATORS

The common idea of derivatives of fractions, mainly the Caputo and Riemann–Liouville[12,13] is discussed.

Definition 1. Derivative with Caputo of $f(x)$ with $\alpha > 0, a, \alpha, x \in$ real numbers is stated as follows:

$$ {}_C D_a^\alpha f(x) = \begin{cases} \dfrac{1}{\Gamma(m-\alpha)} \displaystyle\int_a^x \dfrac{df^{(m)}(t)}{dt^{(m)}} \dfrac{dt}{(x-t)^{\alpha+1-m}}, & m-1 < \alpha \le m \in \mathbb{N}, \\ \dfrac{df^{(m)}(t)}{dt^{(m)}}, & \alpha = m \in \mathbb{N}. \end{cases} \tag{1}$$

The above definition of fractional derivative is important as it holds the ordinary or classic derivative properties as it forms zero of the constant value.[12] For our purposes, we put $m = 1$.

Definition 2. To achieve this purpose, we repeat the fractional derivative of Riemann–Liouville which is the first type of $f(x)$ order α, with $0 < \alpha \le 1$

$$ D_{a+}^\alpha = \begin{cases} \dfrac{1}{\Gamma(1-\alpha)} \dfrac{d}{dx} \displaystyle\int_a^x \dfrac{f(x)}{(x-t)^\alpha} dt, & 0 < \alpha \le 1, \\ \dfrac{df(t)}{dt}, & \alpha = 1. \end{cases} \tag{2}$$

Furthermore, to allow the converging result, now it is essential to enlarge in the Taylor series the $f(x)$ which is not linear using fractional derivatives of Riemann–Liouville.[13]

Definition 3. Caputo and Fabrizio presented a definition of the fractional derivative with a nonsingular kernel, for a function $f(t)$ of C^1 given by

$$ {}_a^{CF} D_t^\alpha f(t) = \frac{1}{1-\alpha} \int_a^t f'(\tau) \exp\left(-\alpha \frac{t-\tau}{1-\alpha}\right) d\tau, \quad \text{where } a < 0, \ \ \alpha \in (0,1]. \tag{3}$$

Definition 4. In mathematical assessment, Newton's strategy, regardless called the Newton–Raphson method, named after Isaac Newton and Joseph Raphson, is a root-discovering calculation that makes constantly better approximations to the roots (or zeroes) of a valid respected cutoff. The most fundamental rendition begins with a solitary variable breaking point f depicted for an affirmed variable x, the cutoff's subordinate f', and a central idea for a foundation of $f(x)$. Assuming the limit satisfies satisfactory doubts and the hidden speculation is close, by then

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

It is anything but a preferable guess of the root over x_0. Mathematically, $(x_1, 0)$ is the convergence of the x-pivot and the digression of the diagram of f at $(x_0, f(x_0))$: that is, the further developed estimate is the special case of the direct guess at the underlying point. The cycle is rehashed as

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

This estimation is first in the class of householder's procedures, won by Halley's technique.

3. CONVERGENCE ANALYSIS OF FRACTIONAL-ORDER NEWTON–RAPHSON METHOD

Theorem 1 (Theorem 3, Refs. 12 and 13). *Assume that* ${}_C D_a^{j\alpha} f(x) \in C([a,b])$ *for* $j = 1, 2, \ldots, n+1$, *as* $\alpha \in (0,1]$, *so is the following*:

$$f(x) = \sum_{i=0}^{n} {}_C D_a^{i\alpha} f(a) \frac{(x-a)^{i\alpha}}{\Gamma(i\alpha+1)} + {}_C D_a^{(n+1)\alpha} f(\xi) \frac{(x-a)^{(n+1)\alpha}}{\Gamma((n+1)\alpha+1)}.$$

With $a \le \xi \le x$, *for all* $x \in (a,b]$ *where* ${}_C D_a^\alpha \cdot {}_C D_a^\alpha \cdots {}_C D_a^\alpha$ *(n times)*.

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Considering the Taylor development of a function $f(x)$ (verifying Theorem 1 hypothesis) at about $a = \bar{x}$ and concerning fractional derivatives of Caputo-type, we have[12]

$$f(x) = \frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}(x-\bar{x})^{\alpha} + \frac{{}_C D_{\bar{x}}^{2\alpha} f(\bar{x})}{\Gamma(2\alpha+1)}(x-\bar{x})^{2\alpha} + \frac{{}_C D_{\bar{x}}^{3\alpha} f(\bar{x})}{\Gamma(3\alpha+1)}(x-\bar{x})^{3\alpha} + O((x-\bar{x})^{4\alpha}).$$

By extracting common $\frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}$, the Taylor development can be represented as follows:

$$f(x) = \frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}[(x-\bar{x})^{\alpha} + C_2(x-\bar{x})^{2\alpha} + C_3(x-\bar{x})^{3\alpha}] + O((x-\bar{x})^{4\alpha}),$$

where $C_j = \frac{\Gamma(\alpha+1)}{\Gamma(j\alpha+1)} \frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}$, for $j \geq 2$.

So, enlargement with the derivative of a fraction of $f(x)$ around $\bar{x}$ is

$${}_C D_a^{\alpha} f(x) = \frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)} \left[\Gamma(\alpha+1) + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 \times (x-\bar{x})^{\alpha} + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 (x-\bar{x})^{2\alpha}\right] + O(x-\bar{x})^{3\alpha}.$$

Hence, by using Caputo fractional derivatives, we have summed up the successive conclusion drawn from the depiction and converging research with the Raphson formula, denoted as CFN.

Theorem 2. *Let*[13] *the continuous $f : D \subseteq \mathbb{R} \to \mathbb{R}$ have fractional derivatives of order $k\alpha$, for any positive integer k and any α, $0 < \alpha \leq 1$, in the interval D containing the zero $\bar{x}$ of $f(x)$. Let us also suppose that ${}_C D_{\bar{x}}^{\alpha} f(\bar{x})$ is continuous and not null in $\bar{x}$. If an initial approximation x_0 is sufficiently close $\bar{x}$, then the local convergence order of the fractional Newton method of Caputo-type*

$$x_{k+1} = x_k - \Gamma(\alpha+1)\frac{f(x_k)}{{}_C D_a^{\alpha} f(x_k)}, \quad k = 0, 1, \ldots$$

is at least 2α, being $0 < \alpha \leq 1$ and the error equation is

$$e_{k+1}^{\alpha} = \frac{\Gamma(2\alpha+1) - (\Gamma(\alpha+1))^2}{(\Gamma(\alpha+1))^3} C_2 e_k^{2\alpha} + O(e_k^{3\alpha}).$$

Theorem 3 (Proposition 3.1, Refs. 14 and 15). *Consider that the continuous function $f : \mathbb{R} \to \mathbb{R}$ being fractional derivatives of order $k\alpha$, then for any positive integer k and any $\alpha, 0 < \alpha \leq 1$, there must be the following equality:*

$$f(x+h) = \sum_{k=0}^{+\infty} \frac{(h)^{k\alpha}}{\Gamma(1+k\alpha)} D_{a+}^{k\alpha} f(x), \quad 0 < \alpha \leq 1,$$

where $D_{a+}^{k\alpha} f(x)$ is a Riemann–Liouville fractional differential of a sequence of $f(x)$.

Therefore, on applied Taylor's expansion which is an identical process since we already define it as Caputo nonordinary differential, we can simply show that the iterative expression is

$$x_{k+1} = x_k - \Gamma(\alpha+1)\frac{f(x_k)}{D_{a+}^{k\alpha} f(x_k)}, \quad k = 0, 1, \ldots.$$

We represent with R-L F N.

Theorem 4. *Let us assume*[13,15] *that continual polynomial $f : D \subseteq \mathbb{R} \to \mathbb{R}$ is nonordinary differential sequence $k\alpha$. There is an integral k with at any $\alpha, 0 < \alpha < 1$, at this interlude D including the root of $\bar{x}$ for function. Also, let us assume that this $D_{a+}^{k\alpha} f(x)$ is continual and also numerous at $\bar{x}$. In case their starting value x_0 is near to $\bar{x}$, their native converging sequence by nonordinary Raphson technique is*

$$x_{k+1} = x_k - \Gamma(\alpha+1)\frac{f(x_k)}{D_{a+}^{k\alpha} f(x_k)}, \quad k = 0, 1, \ldots$$

with Riemann–Liouville kind no less than 2α, as actual $0 < \alpha \leq 1$, then we write once more the formation of error as

$$e_{k+1}^{\alpha} = \frac{\Gamma(2\alpha+1) - (\Gamma(\alpha+1))^2}{(\Gamma(\alpha+1))^3} C_2 e_k^{2\alpha} + O(e_k^{3\alpha}).$$

Further, main feature of quantitative approximation for zeros is that it depends on the starting approximation after convergence. Its feature must examine its successive part.

4. EXAMPLE 1 (POLYNOMIALS FUNCTION)

In this section, we construct the convergence analysis for some examples with the proposed fractional derivatives on Newton–Raphson method.

4.1. Convergence Analysis of Newton-Type Method

This section deals with the convergence analysis of Newton-type method [by applying Theorem 1].

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Here, we consider the polynomial $f(x) = x^3 - 2x - 5$.

$$x^3 - 2x - 5 = \sum_{i=0}^{n} {}_C D_a^{i\alpha}(a^3 - 2a - 5)\frac{(x-a)^{i\alpha}}{\Gamma(i\alpha+1)} + {}_C D_a^{(n+1)\alpha}(\xi^3 - 2\xi - 5) \times \frac{(x-a)^{(n+1)\alpha}}{\Gamma((n+1)\alpha+1)}.$$

With $a \le \xi \le x$, for all $x \in (a, b]$ where ${}_C D_a^{\alpha} \cdot {}_C D_a^{\alpha} \cdots {}_C D_a^{\alpha}$ (n times).

Let us consider the Taylor development of a function $f(x)$ (verifying Theorem 1 hypothesis) at about $a = \bar{x}$, regarding fractional derivatives of Caputo-type[13]

$$x^3 - 2x - 5 = \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3 - 2\bar{x} - 5)}{\Gamma(\alpha+1)}(x-\bar{x})^{\alpha} + \frac{{}_C D_{\bar{x}}^{2\alpha}(\bar{x}^3 - 2\bar{x} - 5)}{\Gamma(2\alpha+1)}(x-\bar{x})^{2\alpha} + \frac{{}_C D_{\bar{x}}^{3\alpha}(\bar{x}^3 - 2\bar{x} - 5)}{\Gamma(3\alpha+1)}(x-\bar{x})^{3\alpha} + O((x-\bar{x})^{4\alpha}).$$

By extracting common $\frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}$, the Taylor development can be represented as follows:

$$(x^3 - 2x - 5) = \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3 - 2\bar{x} - 5)}{\Gamma(\alpha+1)}[(x-\bar{x})^{\alpha} + C_2(x-\bar{x})^{2\alpha} + C_3(x-\bar{x})^{3\alpha}] + O((x-\bar{x})^{4\alpha}),$$

where $C_j = \frac{\Gamma(\alpha+1)}{\Gamma(j\alpha+1)} \frac{{}_C D_{\bar{x}}^{j\alpha}(\bar{x}^3 - 2\bar{x} - 5)}{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3 - 2\bar{x} - 5)}$, for $j \ge 2$.

Then, their parallel enlargement by taking nonordinary derivatives for $(x^3 - 2x - 5)$ about $\bar{x}$ is as

$${}_C D_a^{\alpha}(x^3 - 2x - 5) = \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3 - 2\bar{x} - 5)}{\Gamma(\alpha+1)}\left[\Gamma(\alpha+1) + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 (x-\bar{x})^{\alpha} + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 (x-\bar{x})^{2\alpha}\right] + O((x-\bar{x})^{3\alpha}).$$

4.2. Convergence Order of F · N-Method of Caputo-Type

This section deals with convergence Order of F · N-method of Caputo-type [by applying Theorem 2] for $f(x) = x^3 - 2x - 5$.

$$\frac{f(x_k)}{{}_C D_a^{\alpha} f(x_k)} = \frac{e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} + O e_k^{3\alpha}}{\Gamma(\alpha+1) + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha} + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha} + O(e_k^{3\alpha})},$$

$$f(x_k) = \frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +] + O e_k^{3\alpha},$$

$${}_C D_a^{\alpha} f(x_k) = \frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}\left[\Gamma(\alpha+1) + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha} + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha}\right] + O(e_k^{3\alpha}),$$

$$x_k^3 - 2x_k - 5 = \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3 - 2\bar{x} - 5)}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +] + O e_k^{3\alpha},$$

$$x_k^3 - 2x_k - 5 = \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +] - 2\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +] - 5\frac{{}_C D_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +] + O e_k^{3\alpha},$$

$$x_k^3 - 2x_k - 5 = \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)} e_k^{\alpha} + \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)} C_2 e_k^{2\alpha} + \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)} C_3 e_k^{3\alpha} - 2\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)} e_k^{\alpha} - 2\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)} C_2 e_k^{2\alpha} - 2\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)} C_3 e_k^{3\alpha} - 5\frac{{}_C D_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)} e_k^{\alpha} - 5\frac{{}_C D_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)} C_2 e_k^{2\alpha} - 5\frac{{}_C D_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)} C_3 e_k^{3\alpha} + O e_k^{3\alpha}, \tag{4}$$

$${}_C D_a^{\alpha}(x_k^3 - 2x_k - 5) = \frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x}^3 - 2\bar{x} - 5)}{\Gamma(\alpha+1)}\left[\Gamma(\alpha+1)\right.$$

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$$+\frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)}C_2e_k^{\alpha}+\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}\Big]$$
$$+O(e_k^{3\alpha}),$$

$$\begin{aligned}
&{}_CD_a^{\alpha}(x_k^3-2x_k-5)\\
&=\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}\Big[\Gamma(\alpha+1)+\frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)}C_2e_k^{\alpha}\\
&\quad+\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}\Big]-2\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\\
&\quad\times\Big[\Gamma(\alpha+1)+\frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)}C_2e_k^{\alpha}\\
&\quad+\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}\Big]+9-5\Big[\Gamma(\alpha+1)\\
&\quad+\frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)}C_2e_k^{\alpha}+\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}\Big]\\
&\quad+O(e_k^{3\alpha}),
\end{aligned}$$

$$\begin{aligned}
&{}_CD_a^{\alpha}(x_k^3-2x_k-5)\\
&=\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}\Gamma(\alpha+1)+\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}\\
&\quad\times\frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)}C_2e_k^{\alpha}\\
&\quad+\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}-2\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\\
&\quad\times\Gamma(\alpha+1)-2\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)}C_2e_k^{\alpha}\\
&\quad-2\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}-5\frac{{}_CD_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}\\
&\quad\times\Gamma(\alpha+1)-5\frac{{}_CD_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}\frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)}C_2e_k^{\alpha}\\
&\quad-5\frac{{}_CD_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}+O(e_k^{3\alpha}),
\end{aligned}$$

$$\begin{aligned}
&{}_CD_a^{\alpha}(x_k^3-2x_k-5)\\
&={}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)+{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)\frac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2}C_2e_k^{\alpha}\\
&\quad+\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}\\
&\quad-2{}_CD_{\bar{x}}^{\alpha}(\bar{x})-2{}_CD_{\bar{x}}^{\alpha}(\bar{x})\frac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2}C_2e_k^{\alpha}\\
&\quad-2\frac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}-5{}_CD_{\bar{x}}^{\alpha}\\
&\quad-5{}_CD_{\bar{x}}^{\alpha}\frac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2}C_2e_k^{\alpha}-5\frac{{}_CD_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}\\
&\quad\times\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}+O(e_k^{3\alpha}).
\end{aligned}\tag{5}$$

Dividing Eq. (4) by Eq. (5), we get

$$\begin{aligned}
&\frac{x_k^3-2x_k-5}{{}_CD_{\bar{x}}^{\alpha}(x_k^3-2x_k-5)}\\
&=\frac{\begin{aligned}&\tfrac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}e_k^{\alpha}+\tfrac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}C_2e_k^{2\alpha}+\tfrac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}\\&\times C_3e_k^{3\alpha}-2\tfrac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}e_k^{\alpha}-2\tfrac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}C_2e_k^{2\alpha}\\&-2\tfrac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}C_3e_k^{3\alpha}-5\tfrac{{}_CD_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}e_k^{\alpha}-5\tfrac{{}_CD_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}\\&\times C_2e_k^{2\alpha}-5\tfrac{{}_CD_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}C_3e_k^{3\alpha}+Oe_k^{3\alpha}\end{aligned}}{\begin{aligned}&{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)+{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)\tfrac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2}C_2e_k^{\alpha}\\&+\tfrac{{}_CD_{\bar{x}}^{\alpha}(\bar{x}^3)}{\Gamma(\alpha+1)}\tfrac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}-2{}_CD_{\bar{x}}^{\alpha}(\bar{x})\\&-2{}_CD_{\bar{x}}^{\alpha}(\bar{x})\tfrac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2}C_2e_k^{\alpha}-2\tfrac{{}_CD_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\tfrac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}\\&\times C_3e_k^{2\alpha}-5{}_CD_{\bar{x}}^{\alpha}-5{}_CD_{\bar{x}}^{\alpha}\tfrac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2}C_2e_k^{\alpha}\\&-5\tfrac{{}_CD_{\bar{x}}^{\alpha}}{\Gamma(\alpha+1)}\tfrac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)}C_3e_k^{2\alpha}+O(e_k^{3\alpha}).\end{aligned}}
\end{aligned}$$

4.3. Solution with Riemann–Liouville Fractional Derivative

Solution with Riemann–Liouville fractional derivative [by applying Theorem 3] is as follows:

$$f(x)=x^3-2x-5.$$

Let us consider that the continuous function $f:\mathbb{R}\to\mathbb{R}$ is fractional derivative of order $k\alpha$, then for any positive integer k and any $\alpha, 0<\alpha\le 1$, there must be the following equality satisfying (Proposition 3.1, Ref. 13):

$$\begin{aligned}
&f\{((x+h)^3-2(x+h)-5)\}\\
&\quad=\sum_{k=0}^{\infty}\frac{(h)^{k\alpha}}{\Gamma(1+k\alpha)}D_{a+}^{k\alpha}(x^3-2x-5),\\
&\qquad\qquad 0<\alpha\le 1,
\end{aligned}$$

where $D_{a+}^{k\alpha}(x^3-2x-5)$ is a Riemann–Liouville nonordinary differential of a sequence of $f(x)$.

Therefore, on applying Taylor's expansion which is an identical process since we already define it

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as Caputo nonordinary differential, we can simply show that the iterative expression can be written as

$$x_{k+1} = x_k - \Gamma(\alpha+1)\frac{(x_k^3 - 2x_k - 5)}{D_{a+}^{k\alpha}(x_k^3 - 2x_k - 5)}, \quad k = 0, 1, \ldots.$$

4.4. Local Convergence Order of the F · N-Method with Riemann–Liouville Type

Local convergence order of the F · N-method with Riemann–Liouville type [by applying Theorem 4] for $f(x) = x^3 - 2x - 5$.

Assume that a polynomial $f : D \subseteq \mathbb{R} \to \mathbb{R}$ is nonordinary differential their sequence $k\alpha$, there is at all integral k with at any $\alpha, 0 < \alpha < 1$, at D including the root of $\bar{x}$ for function, assume this $D_{a+}^{k\alpha} f(x)$ is at $\bar{x}$. In case their starting value x_0 is near to $\bar{x}$, so their native converging sequence by nonordinary Raphson technique as

$$x_{k+1} = x_k - \Gamma(\alpha+1)\frac{(x_k^3 - 2x_k - 5)}{D_{a+}^{k\alpha}(x_k^3 - 2x_k - 5)}, \quad k = 0, 1, \ldots$$

with Riemann–Liouville kind no less than 2α, as actual $0 < \alpha \leq 1$, then we write once more the formation of error as

$$e_{k+1}^{\alpha} = \frac{\Gamma(2\alpha+1) - (\Gamma(\alpha+1))^2}{(\Gamma(\alpha+1))^3} C_2 e_k^{2\alpha} + O(e_k^{3\alpha}).$$

Once their converging sequence is formed, its main feature of quantitative approximation for zeros depends on the starting approximation. Its feature must examine its successive part.

5. EXAMPLE-2 (EXPONENTIAL FUNCTION)

5.1. Convergence Analysis of Newton-Type Method

Convergence analysis of Newton-type method [by applying Theorem 1]. Here, we consider the exponential function, i.e. $f(x) = e^x - 3x$ solution with the F · N method.

$$e^x - 3x = \sum_{i=0}^{n} {}_C D_a^{i\alpha}(e^a - 3a)\frac{(x-a)^{i\alpha}}{\Gamma(i\alpha+1)} + {}_C D_a^{(n+1)\alpha}(e^{\xi} - 3\xi)\frac{(x-a)^{(n+1)\alpha}}{\Gamma((n+1)\alpha+1)}.$$

With $a \leq \xi \leq x$, for all $x \in (a, b]$ where ${}_C D_a^{\alpha}{}_C D_a^{\alpha} \cdots {}_C D_a^{\alpha}$ (n times).

Let us consider the Taylor development of a function $f(x)$ (verifying Theorem 1 hypothesis) at about $a = \bar{x}$ and concerning fractional derivatives of Caputo type, we have[13]

$$\begin{aligned} e^x - 3x = {} & \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}} - 3\bar{x})}{\Gamma(\alpha+1)}(x-\bar{x})^{\alpha} \\ & + \frac{{}_C D_{\bar{x}}^{2\alpha}(e^{\bar{x}} - 3\bar{x})}{\Gamma(2\alpha+1)}(x-\bar{x})^{2\alpha} \\ & + \frac{{}_C D_{\bar{x}}^{3\alpha}(e^{\bar{x}} - 3\bar{x})}{\Gamma(3\alpha+1)}(x-\bar{x})^{3\alpha} \\ & + O((x-\bar{x})^{4\alpha}). \end{aligned}$$

By extracting common $\frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}$, the Taylor development can be represented as follows:

$$\begin{aligned} e^x - 3x = {} & \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}} - 3\bar{x})}{\Gamma(\alpha+1)}[(x-\bar{x})^{\alpha} \\ & + C_2(x-\bar{x})^{2\alpha} + C_3(x-\bar{x})^{3\alpha}] \\ & + O((x-\bar{x})^{4\alpha}), \end{aligned}$$

where $C_j = \frac{\Gamma(\alpha+1)}{\Gamma(j\alpha+1)} \frac{{}_C D_{\bar{x}}^{j\alpha}(e^{\bar{x}} - 3\bar{x})}{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}} - 3\bar{x})}$, for $j \geq 2$. So, their parallel enlargement by taking nonordinary derivatives for $(e^x - 3x)$ about $\bar{x}$ is as

$$\begin{aligned} & {}_C D_a^{\alpha}(e^x - 3x) \\ & = \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}} - 3\bar{x})}{\Gamma(\alpha+1)} \left[\Gamma(\alpha+1) + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 \right. \\ & \quad \left. \times (x-\bar{x})^{\alpha} + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 (x-\bar{x})^{2\alpha}\right] \\ & \quad + O((x-\bar{x})^{3\alpha}). \end{aligned}$$

5.2. Convergence Order of F · N-Method of Caputo-Type

Convergence order of F · N-method of Caputo-type [by applying Theorem 2] for $f(x) = e^x - 3x$.

$$\frac{f(x_k)}{{}_C D_a^{\alpha} f(x_k)} = \frac{e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} + O e_k^{3\alpha}}{\Gamma(\alpha+1) + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha} + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha} + O(e_k^{3\alpha})},$$

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$$
\begin{aligned}
&f(x_k) = \frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +] + O e_k^{3\alpha},\\
&{}_C D_a^{\alpha} f(x_k)\\
&\quad = \frac{{}_C D_{\bar{x}}^{\alpha} f(\bar{x})}{\Gamma(\alpha+1)}\left[\Gamma(\alpha+1) + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha}\right.\\
&\qquad \left. + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha}\right] + O(e_k^{3\alpha}),\\
&e^{x} - 3x_k\\
&\quad = \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}} - 3\bar{x})}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +]\\
&\qquad + O e_k^{3\alpha},\\
&e^{x_k} - 3x_k\\
&\quad = \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +]\\
&\qquad - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}[e_k^{\alpha} + C_2 e_k^{2\alpha} + C_3 e_k^{3\alpha} +]\\
&\qquad + O e_k^{3\alpha},\\
&e^{x_k} - 3x_k\\
&\quad = \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)} e_k^{\alpha} + \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)} C_2 e_k^{2\alpha}\\
&\qquad + \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)} C_3 e_k^{3\alpha} - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)} e_k^{\alpha}\\
&\qquad - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)} C_2 e_k^{2\alpha} - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\\
&\qquad \times C_3 e_k^{3\alpha} + O e_k^{3\alpha},
\end{aligned}
\tag{6}
$$

$$
\begin{aligned}
&{}_C D_a^{\alpha}(e^{x_k} - 3x_k)\\
&\quad = \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}} - 3\bar{x})}{\Gamma(\alpha+1)}\left[\Gamma(\alpha+1)\right.\\
&\qquad \left. + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha} + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha}\right]\\
&\qquad + O(e_k^{3\alpha}),\\
&{}_C D_a^{\alpha}(e^{x_k} - 3x_k)\\
&\quad = \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)}\left[\Gamma(\alpha+1)\right.\\
&\qquad \left. + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha} + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha}\right]\\
&\qquad - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\left[\Gamma(\alpha+1) + \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha}\right.\\
&\qquad \left. + \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha}\right] + O(e_k^{3\alpha}),\\
&{}_C D_a^{\alpha}(e^{x_k} - 3x_k)\\
&\quad = \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)}\Gamma(\alpha+1) + \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)}\\
&\qquad \times \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha} + \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)}\\
&\qquad \times \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha} - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\\
&\qquad \times \Gamma(\alpha+1) - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\\
&\qquad \times \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)} C_2 e_k^{\alpha} - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\\
&\qquad \times \frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha} + O(e_k^{3\alpha}),\\
&{}_C D_a^{\alpha}(e^{x_k} - 3x_k)\\
&\quad = {}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}}) + {}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})\frac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2} C_2 e_k^{\alpha}\\
&\qquad + \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha}\\
&\qquad - 3{}_C D_{\bar{x}}^{\alpha}(\bar{x}) - 3{}_C D_{\bar{x}}^{\alpha}(\bar{x})\frac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2} C_2 e_k^{\alpha}\\
&\qquad - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha} + O(e_k^{3\alpha}).
\end{aligned}
\tag{7}
$$

Dividing Eq. (6) by Eq. (7), we get

$$
\begin{aligned}
&\frac{e^{x_k} - 3x_k}{{}_C D_a^{\alpha}(e^{x_k} - 3x_k)}\\
&\quad = \frac{\begin{array}{l}\frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)} e_k^{\alpha} + \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)} C_2 e_k^{2\alpha} + \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)}\\ \quad \times C_3 e_k^{3\alpha} - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)} e_k^{\alpha} - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)} C_2 e_k^{2\alpha}\\ \quad - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)} C_3 e_k^{3\alpha} + O e_k^{3\alpha}\end{array}}{\begin{array}{l}{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}}) + {}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})\frac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2} C_2 e_k^{\alpha}\\ \quad + \frac{{}_C D_{\bar{x}}^{\alpha}(e^{\bar{x}})}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha} - 3{}_C D_{\bar{x}}^{\alpha}(\bar{x})\\ \quad - 3{}_C D_{\bar{x}}^{\alpha}(\bar{x})\frac{\Gamma(2\alpha+1)}{(\Gamma(\alpha+1))^2} C_2 e_k^{\alpha}\\ \quad - 3\frac{{}_C D_{\bar{x}}^{\alpha}(\bar{x})}{\Gamma(\alpha+1)}\frac{\Gamma(3\alpha+1)}{\Gamma(2\alpha+1)} C_3 e_k^{2\alpha} + O(e_k^{3\alpha})\end{array}}
\end{aligned}
$$

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5.3. Solution with Riemann–Liouville Fractional Derivative

Solution with Riemann–Liouville fractional derivative [by applying Theorem 3] for

$$f(x) = e^x - 3x.$$

Consider that the continuous function $f : \mathbb{R} \to \mathbb{R}$ being fractional derivatives of order $k\alpha$, then for any positive integer k and any $\alpha, 0 < \alpha \leq 1$, there must be the following equality (Proposition 3.1, Ref. 13):

$$f\{(e^{x+h} - 3(x+h))\} = \sum_{k=0}^{\infty} \frac{(h)^{k\alpha}}{\Gamma(1+k\alpha)} D_{a+}^{k\alpha}(e^x - 3x), \quad 0 < \alpha \leq 1,$$

where $D_{a+}^{k\alpha}(e^x - 3x)$ is a Riemann–Liouville nonordinary differential of a sequence of $f(x)$.

Table 1 Fractional Newton Conclusions for $f(x)$ and Starting Approximation $x_0 = 2.1$. Table with CFN Method.

α	$\bar{x}$	$\lvert x_{k+1} - x_k \rvert$	$\lvert f(x_{k+1}) \rvert$	Iter
0.7	2.09455148154233	2.09999891322	$3.5527136788005e-15$	500
0.72	2.09455148154233	0	$3.5527136788005e-15$	500
0.74	2.09455148154233	0	$3.5527136788005e-15$	500
0.76	2.09455148154233	0	$3.5527136788005e-15$	500
0.78	2.09455148154233	0	$3.5527136788005e-15$	500
0.8	2.09455148154233	0	$3.5527136788005e-15$	500
0.82	2.09455148154233	0	$3.5527136788005e-15$	500
0.84	2.09455148154233	0	$3.5527136788005e-15$	500
0.86	2.09455148154233	0	$3.5527136788005e-15$	500
0.88	2.09455148154233	0	$3.5527136788005e-15$	500
0.9	2.09455148154233	0	$3.5527136788005e-15$	500
0.92	2.09455148154233	0	$3.5527136788005e-15$	500
0.94	2.09455148154233	0	$3.5527136788005e-15$	500
0.96	2.09455148154233	0	$3.5527136788005e-15$	500
0.98	2.09455148154233	0	$3.5527136788005e-15$	500
1	2.09455148154233	0	$3.5527136788005e-15$	500

Table 2 Fractional Newton Conclusions for $f(x)$ and Starting Approximation $x_0 = 2.1$. Table with R-LFN Method

α	$\bar{x}$	$\lvert x_{k+1} - x_k \rvert$	$\lvert f(x_{k+1}) \rvert$	Iter
0.7	2.09455148154233	2.09999891322	$3.5527136788005e-15$	500
0.72	2.09455148154233	0	$3.5527136788005e-15$	500
0.74	2.09455148154233	0	$3.5527136788005e-15$	500
0.76	2.09455148154233	0	$3.5527136788005e-15$	500
0.78	2.09455148154233	0	$3.5527136788005e-15$	500
0.8	2.09455148154233	0	$3.5527136788005e-15$	500
0.82	2.09455148154233	0	$3.5527136788005e-15$	500
0.84	2.09455148154233	0	$3.5527136788005e-15$	500
0.86	2.09455148154233	0	$3.5527136788005e-15$	500
0.88	2.09455148154233	0	$3.5527136788005e-15$	500
0.9	2.09455148154233	0	$3.5527136788005e-15$	500
0.92	2.09455148154233	0	$3.5527136788005e-15$	500
0.94	2.09455148154233	0	$3.5527136788005e-15$	500
0.96	2.09455148154233	0	$3.5527136788005e-15$	500
0.98	2.09455148154233	0	$3.5527136788005e-15$	500
1	2.09455148154233	0	$3.5527136788005e-15$	500

Therefore, by applying Taylor's expansion which is an identical process since we already define it as Caputo nonordinary differential, we can simply show that the iterative expression is

$$x_{k+1} = x_k - \Gamma(\alpha+1)\frac{(e^{x_k} - 3x_k)}{D_{a+}^{k\alpha}(e^{x_k} - 3x_k)}, \quad k = 0, 1, \ldots.$$

5.4. Local Convergence Order of the F · N-Method with Riemann–Liouville Type

Local convergence order of the F. N-method with Riemann–Liouville type [by applying Theorem 4] for $f(x) = e^x - 3x$.

Assume that continual polynomial $f : D \subseteq \mathbb{R} \to \mathbb{R}$ is nonordinary differential their sequence $k\alpha$,

Table 3 Fractional Newton Conclusions for $f(x)$ and Starting Approximation $x_0 = 0$. Table with CFN Method.

α	$\bar{x}$	$\lvert x_{k+1} - x_k \rvert$	$\lvert f(x_{k+1}) \rvert$	iter
0.7	0.619061286735945	0	0	500
0.72	0.619061286735945	0	0	500
0.74	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.76	0.619061286735945	$2.22044604925031e-16$	0	500
0.78	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.8	0.619061286735945	$2.22044604925031e-16$	0	500
0.82	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.84	0.619061286735945	0	$2.22044604925031e-16$	500
0.86	0.619061286735945	$2.22044604925031e-16$	0	500
0.88	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.9	0.619061286735945	$2.22044604925031e-16$	0	500
0.92	0.619061286735945	0	0	500
0.94	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.96	0.619061286735945	0	$2.22044604925031e-16$	500
0.98	0.619061286735945	0	$2.22044604925031e-16$	500
1	0.619061286735945	$2.22044604925031e-16$	0	500

Table 4 Fractional Newton Conclusions for $f(x)$ and Starting Approximation $x_0 = 0$. Table with R-LFN Method.

α	$\bar{x}$	$\lvert x_{k+1} - x_k \rvert$	$\lvert f(x_{k+1}) \rvert$	iter
0.7	0.619061286735945	0	0	500
0.72	0.619061286735945	0	0	500
0.74	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.76	0.619061286735945	$2.22044604925031e-16$	0	500
0.78	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.8	0.619061286735945	$2.22044604925031e-16$	0	500
0.82	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.84	0.619061286735945	0	$2.22044604925031e-16$	500
0.86	0.619061286735945	$2.22044604925031e-16$	0	500
0.88	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.9	0.619061286735945	$2.22044604925031e-16$	0	500
0.92	0.619061286735945	0	0	500
0.94	0.619061286735945	$2.22044604925031e-16$	$2.22044604925031e-16$	500
0.96	0.619061286735945	0	$2.22044604925031e-16$	500
0.98	0.619061286735945	0	$2.22044604925031e-16$	500
1	0.619061286735945	$2.22044604925031e-16$	0	500

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there is at all integral k with at any $\alpha, 0 < \alpha < 1$, at this interlude D including the root of $\bar{x}$ for function. Along assume this $D_{a+}^{k\alpha} f(x)$ is continual also numerous at $\bar{x}$. In case their starting value x_0 is near to $\bar{x}$, so their native converging sequence by nonordinary Raphson technique as

$$x_{k+1} = x_k - \Gamma(\alpha+1)\frac{(e^{x_k} - 3x_k)}{D_{a+}^{k\alpha}(e^{x_k} - 3x_k)}, \quad k = 0, 1, \ldots$$

with Riemann–Liouville kind no less than 2α, as actual $0 < \alpha \leq 1$, then we write once more the formation of error as

$$e_{k+1}^{\alpha} = \frac{\Gamma(2\alpha+1) - (\Gamma(\alpha+1))^2}{(\Gamma(\alpha+1))^3} C_2 e_k^{2\alpha} + O(e_k^{3\alpha}).$$

Once their converging sequence is formed, further main feature of quantitative approximation for zeros depends on the starting approximation. Its feature must examine its successive part.

6. NUMERICAL RESULTS AND DISCUSSION

Here, we have a polynomial

$$f(x) = x^3 - 2x - 5.$$

Now again by using two fractional techniques, conclusions can be drawn.

Here, we have a function, $f(x) = e^x - 3x$.

Now again by using two fractional techniques, conclusions can be drawn as under. Results are presented by Tables 1–4.

7. CONCLUSION

In this work, we proposed the Newton–Raphson Method with Riemann–Liouville and Caputo fractional derivatives. We consider the polynomial, exponential function, and trigonometric functions for observation of result utilization of the proposed scheme. It is important to use a Gamma function as a damping boundary to demonstrate the 2αth convergence of the two strategies. Some mathematical tests were prepared and the reliance on the underlying assessment on envisioned as intermingling planes. In them, various exhibitions were discovered when both partial subordinates were utilized, regardless of whether the stringently mathematical outcomes were comparable. Consequently, we observed that CFN provides a better result as far as the largeness of the bowls of fascinating roots with the autonomy of the worth of α. The fractional-order Newton–Raphson method can also be used to solve the power flow problem. Only the polar form is employed in practice because it generates fewer equations than the rectangular form, which results in a larger number of equations. By using fractional-order techniques, we can find roots in the complex plane as well other than real roots for biological, engineering, and social issues.

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